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CASE REPORT

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Efficacy of a personalized combination regimen combining induction chemotherapy, immunotherapy, radiotherapy, and targeted therapy for locally advanced synchronous multiple primary lung cancer: a case report

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Abstract

Background Multiple primary lung cancer is becoming increasingly common in clinical practice. It is divided into synchronous multiple primary lung cancer and nonsynchronous multiple primary lung cancer on the basis of a diagnostic interval of 6 months. However, the diagnosis and treatment of multiple primary lung cancer face great challenges in clinical practice.

Case presentation

We report a case of a 66-year-old Chinese male patient with locally advanced synchronous multifocal lung cancer. In M 0 months (the time of initial diagnosis), the patient underwent two cycles of carboplatin combined with albumin-bound paclitaxel induction chemotherapy before treatment was suspended due to personal reasons. In M+10 months, he resumed a personalized multimodal treatment regimen: induction chemotherapy combined with immunotherapy, concurrent chemoradiotherapy combined with targeted therapy, and adjuvant immunotherapy combined with targeted maintenance therapy. As of M+72 months, positron positron emission tomography–computed tomography showed lung metastasis, but his Karnofsky Performance Status score remained at 80, and he achieved progression-free survival of 72 months. He continues to receive chemotherapy, tislelizumab immunotherapy, and recombinant human endothelin (Endostar) targeted therapy. As of M+75 months, he has survived for 75 months without reaching the overall survival endpoint.

Conclusion Clinical practice should adopt a multidisciplinary treatment-based approach, integrating clinical information, imaging data, histopathological features, and molecular characteristics to formulate individualized clinical management strategies and treatment plans. Such personalized therapeutic approaches are essential to improving long-term survival rates and quality of life in patients with synchronous multifocal lung cancer.

Keywords Multiple primary lung cancer, Chemotherapy, Radiotherapy, Immunotherapy, Targeted therapy

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Introduction

Multiple primary lung cancer (MPLC) is a distinct and complex clinical entity characterized by the presence of two or more independent primary lung tumors without evidence of metastasis [1]. As early as 1924, Beyreuther reported the incidence of MPLC to range from 0.8% to 14.5%. Studies have shown that sMPLC is more common, predominantly occurring in adenocarcinoma, but relatively rare in squamous cell carcinoma [2]. Currently, sMPLC is defined as the presence of two or more primary lung cancer lesions diagnosed simultaneously or within 6 months in the same patient. In contrast, metachronous MPLC (mMPLC) refers to a new primary lung lesion detected more than 6 months after initial lung cancer treatment [3]. The current clinical challenge lies in distinguishing between independent primary lung cancer (IPLC) and intra-pulmonary metastases (IPM) in patients with multiple pulmonary cancers, which is critical for staging, treatment decisions, and prognosis assessment. Compared with MPLC, IPM typically results in higher staging and poorer prognosis. The 8th edition of the International Association for the Study of Lung Cancer (IASLC) TNM Staging Manual classifies IPMs as T3 (same lobe), T4 (different lobe), or M1a (contralateral lung), while sMPLC are staged individually for each tumor [4]. Although the clinical community has made significant efforts to explore the diagnosis and treatment of synchronous multiple primary lung cancer (sMPLC), its potential pathophysiological mechanisms are still largely unknown, and distinguishing sMPLC from intrapulmonary metastasis (IPM) remains challenging. The NCCN and ESMO have not yet established standardised treatment protocols for sMPLC, and current management follows the principles for nonsmall cell lung cancer (NSCLC): For R0 resection that can be completed in a single procedure and where lung function permits, surgical treatment remains the preferred option (lobectomy/segmentectomy ± resection). For locally advanced stages (T3-T4 or N2), surgical feasibility should be prioritized for assessment; patients ineligible for surgical resection will receive concurrent radiotherapy (cisplatin-based dual-agent regimen at 60–66 Gy), followed by 12 months of durvalumab therapy (PACIFIC regimen). Unresectable or oligometastatic NSCLC should be treated as metastatic NSCLC: Patients with PD-L1 ≥ 50% and no driver gene mutations should receive pembrolizumab monotherapy; for patients with driver gene mutations, targeted sequential therapy should be applied as appropriate. The ESMO 2023 consensus states that nonsurgical patients may receive personalized stereotactic body radiation therapy (SBRT); if mediastinal lymph node metastasis is present, curative concurrent chemotherapy and radiotherapy should be performed. In summary, current guidelines still follow the criteria of resectability and molecular/PD-L1 status, adhere to the treatment regimen for single NSCLC, and emphasize personalized decision-making by a multidisciplinary team (MDT). This case report aims to explore potential treatment approaches that may benefit patients with sMPLC.

Case presentation

A 66-year-old Chinese male patient was admitted to our hospital with a chief complaint of "over 3 years of comprehensive treatment for squamous cell carcinoma (SCC) of both lungs, with a newly developed lesion in the left lung for 20 days." In M 0 months (the time of initial diagnosis), he experienced intermittent hemoptysis (approximately 4 mL of fresh blood each time). Chest computed tomography (CT) with contrast enhancement performed revealed a mass in the right hilum with associated hilar and mediastinal lymphadenopathy, suggestive of lung cancer (* MERGEFORMAT Fig. 1). Pathological biopsy confirmed SCC; however, the patient did not undergo further standardized evaluation or treatment. In M + 3 months, the patient presented again with cough and hemoptysis at the Hospital. Tumor marker testing showed elevated cytokeratin (5.14 ng/mL↑). On the basis of the systemic evaluation, the patient was clinically diagnosed with SCC of the right upper lung (cT4N2M0, stage IIIb), with a Karnofsky Performance Status (KPS) score of 80. In M + 4 months, he subsequently received two cycles of neoadjuvant chemotherapy with carboplatin and albumin-bound paclitaxel. After chemotherapy, re-examination by chest CT showed stable disease (SD) (Fig. 1). Further improve the inspection, A bronchoscopy revealed a neoplasm obstructing the right bronchus and another smooth-surfaced lesion in the left bronchus. Biopsy of both lesions confirmed SCC, with immunohistochemical staining positive for p40 and p63 (Fig. 2). Peripheral blood genetic testing showed PD-L1 expression of 20%, with no other driver mutations detected. According to the Martini-Melamed criteria, multiple pulmonary lesions with identical histology located in different lobes or lungs without common lymphatic spread or distant metastasis may be diagnosed as sMPLC. Although the initial CT scan showed mediastinal lymphadenopathy, the right and left lung lesions exhibited distinct lymphatic metastasis patterns (right side N2 vs. left side N0) without involvement of shared lymphatic drainage areas. After multidisciplinary team (MDT) discussion, the diagnosis favored sMPLC. Thus, the final diagnosis was SCC of the right upper lung (ycT4N2M0, stage IIIb) and SCC of the left upper lung (ycT1N0M0, stage Ia). However, the patient declined further recommended treatment and self-administered traditional Chinese medicine at home.

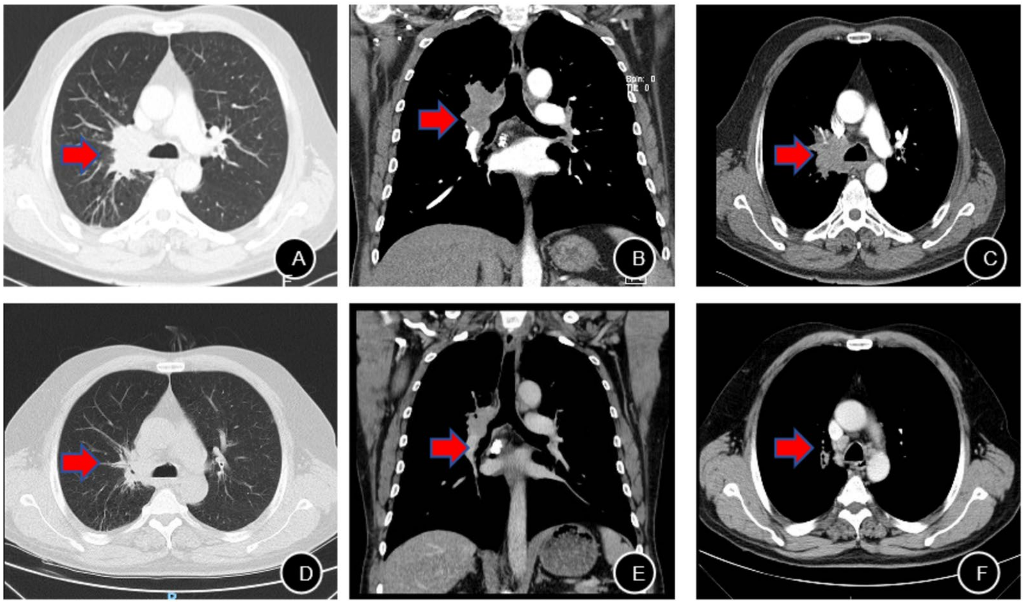

Fig. 1 Chest contrast-enhanced and plain computed tomography scans In M0 months: **A**, **B** red arrow indicates a soft tissue mass in the right hilum measuring approximately 4.5 cm × 3.1 cm × 4.8 cm, with truncation of the right upper lobe bronchus and obstructive pneumonia in the right upper lobe; **C** Multiple enlarged lymph nodes in the mediastinum and right hilum. The red arrow suggests of lymph node metastasis. M + 5 months two cycles of neoadjuvant chemotherapy Re-examination Chest contrast-enhanced and plain computed tomography scans: **D**, **E** A soft tissue density mass in the right hilar region measuring approximately 3.9 cm × 2.8 cm × 4.1 cm. Red arrow indicates reduction in lung cancer volume; **F** red arrow indicates reduced volume of multiple enlarged lymph nodes in the right hilum and mediastinum

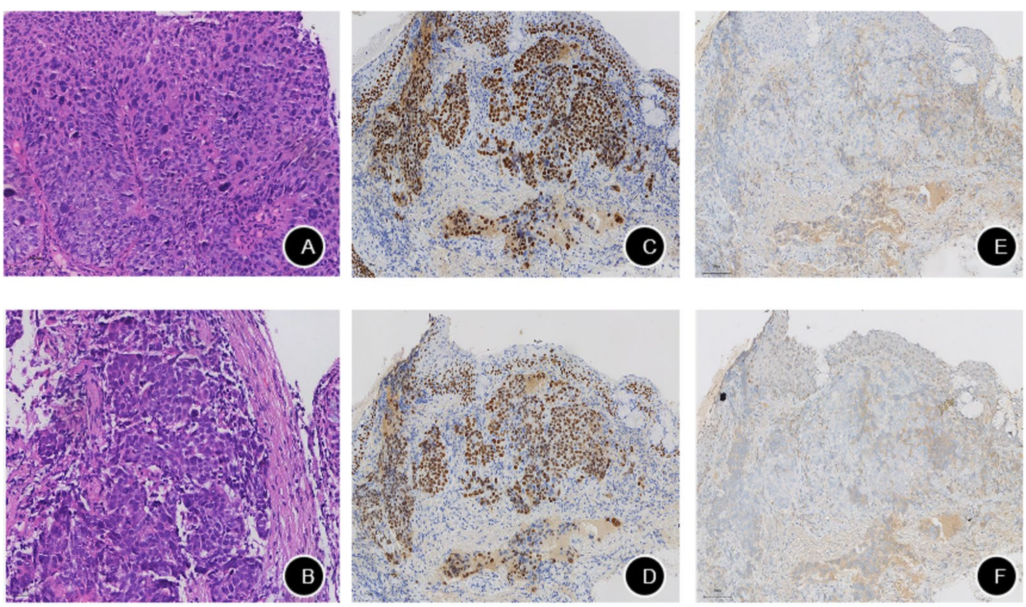

Fig. 2 Histopathological and immunohistochemical staining of bilateral lung biopsy specimens under the microscope. **A** Invasive growth of tumor cells infiltrating the right lung (H&E staining × 200); **B** Invasive growth of tumor cells infiltrating the left lung (H&E staining × 200); **C** Positive P40 expression; **D** Positive P63 expression; **E** Negative NapsinA expression; **F** Negative TTF-1 expression

In M + 9 months, positron emission tomography–computed tomography (PET–CT) re-examination showed enlargement of the right hilar lesion with newly developed mediastinal lymph node metastases (Fig. 3). At M + 10 months, the patient presented to the hospital. On the basis of clinical evaluation and MDT discussion, the diagnosis of SCC of the right upper lung (ycT4N2M0, stage IIIb) and the left lung (ycT1N0M0, stage Ia) was confirmed. The patient has a history of smoking and chronic obstructive pulmonary disease (COPD), and PET–CT imaging revealed obstructive pneumonia. Surgical assessment indicated poor lung function and high surgical risk. Considering the patient's physical condition and treatment-related toxicity, curative surgery and concurrent radiotherapy were deemed inappropriate. At the same time, considering the patient's advanced age and extensive tumor burden, which precluded radical surgery or concurrent chemoradiotherapy, a personalized, comprehensive treatment strategy was recommended.

The patient subsequently received two cycles of induction therapy with carboplatin plus albumin-bound paclitaxel combined with Toripalimab in M + 10–11 months. During induction therapy, the patient developed recurrent grade III myelosuppression, primarily manifesting as neutropenia and thrombocytopenia. Upon completion of induction therapy, radiotherapy was initiated in M + 13 months using 6-MVX beams. The prescribed dose was 60 Gy delivered to the planning target volume (PTV) in 30 fractions, encompassing the residual right lung lesion post-chemotherapy and high-risk lymphatic drainage regions. To enhance treatment efficacy, the patient concurrently received two cycles of recombinant human endostatin (Endostar) as anti-angiogenic targeted therapy during radiotherapy. Following the completion of radiotherapy, the patient developed hypothyroidism, which was managed by endocrinology specialists, with normalization of thyroid function upon follow-up. Thereafter, in M + 15–72 months, the patient underwent 19 cycles of maintenance therapy combining immunotherapy with anti-angiogenic therapy, consisting of pembrolizumab 200 mg on day 1 and Endostar 210 mg continuously from day 1 to day 14 of each cycle.

During regular follow-up, the PET–CT scan in M + 24 months demonstrated stable disease (Fig. 3). However, in M + 72 months, PET–CT revealed increased metabolic activity in a nodule within the apical-posterior segment of the left upper lobe, suggesting disease recurrence. Additionally, multiple new small nodules with elevated metabolic activity were detected in both lungs, which is highly suspicious for metastatic disease (Fig. 4). Serum CYFRA 21–1 was elevated to 3.64 ng/mL. Bronchoscopic biopsy of the left lung lesion confirmed squamous cell carcinoma, with immunohistochemistry showing CK7 (partial positive), TTF-1 (negative), Napsin A (negative), P40 (positive), and P63 (positive). On the basis of the patient's medical history and auxiliary

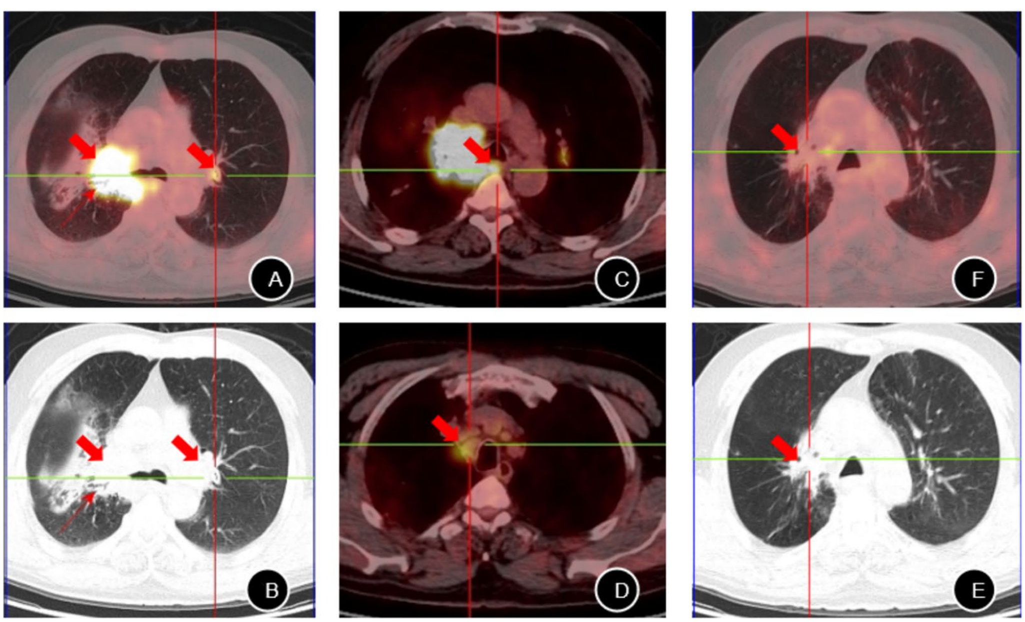

Fig. 3 Whole-body positron emission tomography–computed tomography images In M + 9 months: Whole-body positron emission tomography–computed tomography images on May 28, 2019: **A**, **B** Red arrow indicates a hypermetabolic mass in the right hilar region measuring approximately 3.85 cm × 4.97 cm × 6.35 cm with a maximum SUV of 15.94; a focal hypermetabolic lesion in the apical-posterior segment of the left upper lobe measuring approximately 0.35 cm with a maximum SUV of 4.56; **C**, **D** Red arrow indicates hypermetabolic lymph nodes in the mediastinal 2R and 3P regions with maximum SUVs ranging from 3.51 to 6.67.Whole-body positron emission tomography–computed tomography images In M + 24 months: **E**, **F** Red arrow indicates a soft tissue density shadow in the right hilar region with low metabolic activity measuring approximately 2.6 cm × 3.6 cm × 2.3 cm, with FDG uptake disappearing compared with the previous scan; no obvious FDG-avid nodules detected in the previous hypermetabolic lesion in the apical-posterior segment of the left upper lobe

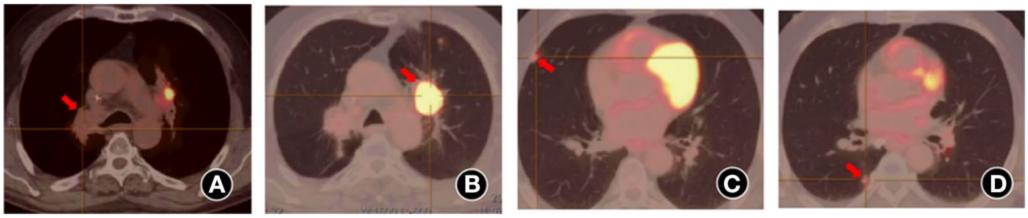

Fig. 4 Whole-body positron emission tomography–computed tomography images in M + 72 months: **A** red arrow indicates a patchy soft tissue density shadow in the right hilar region measuring approximately 2.6 cm × 3.2 cm × 2.5 cm with a maximum SUV of 2.34, indicating suppressed tumor activity after treatment; **B** Red arrow indicates a nodular lesion adjacent to the aorta in the apical-posterior segment of the left upper lobe measuring approximately 3.3 cm × 2.7 cm × 3.7 cm with a maximum SUV of 6.09–17.31, suggestive of tumor recurrence; **C**, **D** Red arrow indicates multiple newly emerged scattered small nodules in various lobes of both lungs, measuring between 0.3 cm × 0.2 cm and 1.1 cm × 0.8 cm, with maximum SUVs ranging from 1.26 to 4.67, indicating possible intrapulmonary metastasis

examinations, the diagnosis was confirmed as recurrent pulmonary squamous cell carcinoma with multiple bilateral lung metastases (rcT4N0M1, stage IVa). A first-line treatment regimen for advanced metastatic lung Squamous Cell Carcinoma Antigen (SCC) was initiated, consisting of chemotherapy combined with tislelizumab immunotherapy and recombinant human endostatin (Endostar) targeted therapy. The specific treatment protocol included tislelizumab 200 mg, paclitaxel 380 mg on day 1, carboplatin 400 mg on day 1, and Endostar 210 mg administered via intravenous pump from day 1 to day 3.

Management and outcome

From M0 months to M + 72 months, the patient underwent a contrast-enhanced chest CT scan every three months to monitor treatment efficacy. PET–CT was used to confirm suspected recurrent lesions, and tumor markers (SCC, CYFRA21-1, and CEA) were checked monthly. In M + 72 months, PET–CT scan revealed multiple lung metastases, with serum CYFRA 21–1 elevated to 3.64 ng/mL. The diagnosis was confirmed as recurrent pulmonary squamous cell carcinoma with multiple bilateral lung metastases (rcT4N0M1, stage IVa). Given the unresectable lung metastases, treatment was recommended with chemotherapy, tislelizumab immunotherapy, and recombinant human endostatin (Endostar). As of M + 75 months, the patient's KPS score remained at 80. He achieved progression-free survival (PFS) of 72 months, survived for 75 months without reaching the overall survival (OS) endpoint (Fig. 5).

Discussion

The diagnosis and treatment of sMPLC remain challenging in clinical practice. In particular, for cases with similar histological subtypes, there is still considerable controversy regarding the diagnostic criteria for sMPLC and its differentiation from intrapulmonary metastasis (IM). According to the Martini-Melamed criteria, sMPLC can be diagnosed if tumors share the same histology but are located in different lung segments, lobes, or lungs without evidence of common lymph node metastasis or distant metastasis [5]. On the basis of the modified Antakli criteria (1995), a diagnosis of sMPLC can also be considered in patients with tumors of identical histology when two or more of the following conditions are met: anatomically separate lesions (located in different lobes), presence of related premalignant lesions; absence of extrathoracic metastasis; absence of mediastinal spread; and different DNA ploidy status between lesions [6]. Currently, the main treatment modalities for MPLC include surgery, radiotherapy, and ablation, while targeted therapy and immunotherapy are generally reserved for selected patients. However, in real-world clinical settings, the number, location, and size of MPLC lesions often influence treatment decision-making, and there is still no consensus on the optimal treatment strategy for sMPLC.

For patients with early-stage sMPLC (including stage I disease and multiple ground-glass/lepidic lesions), surgical treatment generally yields favorable outcomes. Nevertheless, prognostic and risk factors for postoperative recurrence in this population have not been clearly defined [7]. Hui Zhang *et al.* conducted a retrospective analysis of 293 patients with sMPLC who underwent surgical resection and reported 10-year overall survival (OS) and recurrence-free survival (RFS) rates of 96.1% and 92.8%, respectively. Their study identified preoperative FEV1 (%), pathological subtype, highest pathological T stage (pT), and lymphovascular invasion (LVI) as key prognostic factors influencing OS and RFS. In addition, the combined tumor size and presence of pleural invasion were significant predictors of RFS [8]. NIE Y *et al.* suggested that sublobar resection is feasible for early-stage sMPLC patients, while pneumonectomy should be avoided to preserve postoperative quality of life. Their study also reported that lymph node metastasis (hazard ratio [HR] = 2.36; 95% confidence interval [CI] 1.75–3.20;

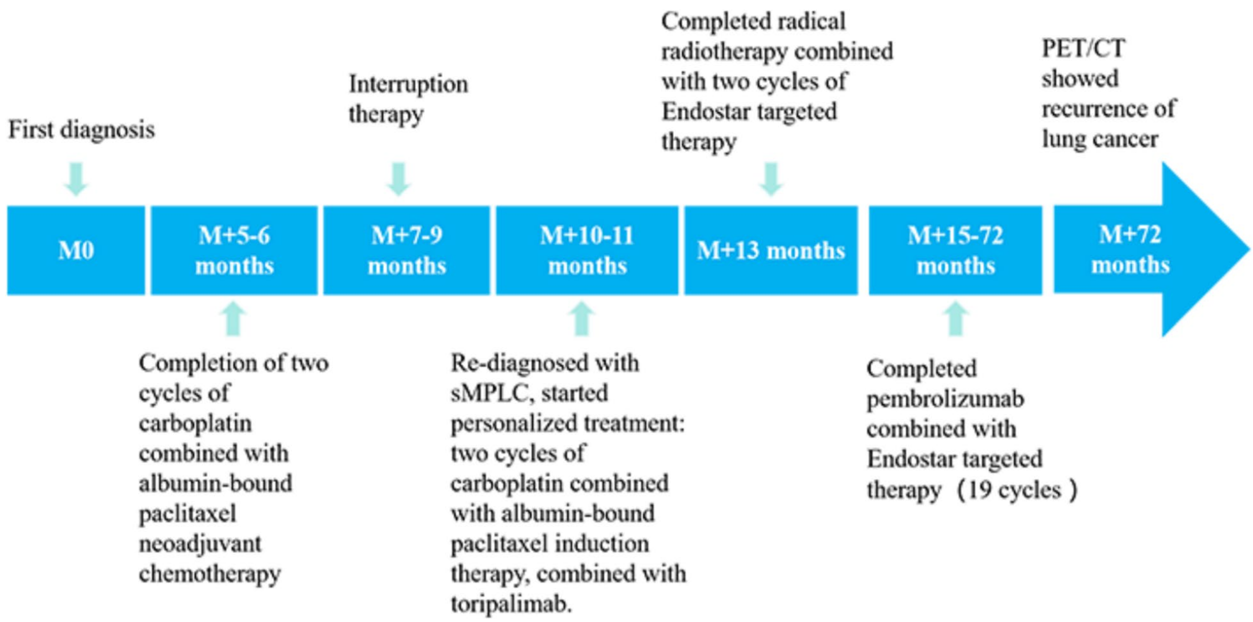

Fig. 5 Timeline of clinical events

$P<0.001$) and pneumonectomy (HR = 2.96; 95% CI 1.36–6.45; $P=0.006$) were adverse prognostic factors for sMPLC [9]. In a study by Yang *et al.*, propensity score matching was used to minimize potential bias caused by clinicopathological differences between the first and second primary lung tumors. The results showed that for tumors with a diameter of ≤ 2 cm, the prognosis of patients undergoing sublobar resection was comparable to that of those undergoing lobectomy. However, for tumors measuring 2–3 cm, patients who received lobectomy had significantly better survival outcomes than those who underwent sublobar resection ($P<0.014$) [10]. In summary, the surgical management of early-stage sMPLC requires comprehensive preoperative evaluation of tumor characteristics and patient condition to determine surgical indications. The extent of resection should be tailored according to lesion location and size, with an appropriate selection of candidates for anatomical resection, to avoid unnecessary surgical trauma and preserve pulmonary function following surgery, particularly in cases where pneumonectomy would otherwise be required [11].

Comprehensive treatment is particularly critical for sMPLC patients who are not candidates for surgical resection. According to the PACIFIC and PACIFIC-R studies, concurrent chemoradiotherapy (CRT) followed by sequential immunotherapy has been recognized as the standard treatment strategy for unresectable stage III NSCLC. In these studies, the 12-month and 18-month PFS rates were 55.9% and 44.2%, respectively, with a median OS of 47.5 months, significantly improving patient prognosis [12, 13]. However, there is substantial heterogeneity among patients with stage III NSCLC in real-world clinical practice. Factors such as poor tolerance to cCRT, advanced age, frailty, comorbidities, tumor volume, and lesion location often lead to the selection of sequential chemoradiotherapy (sCRT) for some patients. A phase II single-arm study explored the efficacy and safety of neoadjuvant immunotherapy prior to definitive chemoradiotherapy in unresectable stage III NSCLC patients. The results demonstrated a median PFS of 30.0 months (95% CI 15.8 months–not reached), with the median OS not yet reached. The 24-month OS rate was 73.7% (95% CI 63.4%–85.7%), and the objective response rate (ORR) was 66.2% [14]. Compared with the PACIFIC study, adding neoadjuvant atezolizumab did not reduce therapeutic efficacy. It may even improve PFS through early initiation of immune checkpoint inhibitor (ICI) therapy, although long-term survival outcomes are still under follow-up.

At the 2024 ASCO Annual Meeting, Arthi Sridhar *et al.* presented data from the PACIFIC study showing that for stage IIIA and IIIB NSCLC patients, the median PFS and OS were 16.9 months and 47.5 months, respectively, but the risk of recurrence remained high. Therefore, they proposed a strategy of induction chemo-immunotherapy (ID chemo-IO) followed by chemoradiotherapy (CRT) [15], enrolling 927 patients with stage III NSCLC, including adenocarcinoma (66.67%), squamous cell carcinoma (27.8%), and other pathological subtypes. Immunohistochemistry (IHC) testing showed PD-L1 expression of 1–49% in 47.2% of patients, > 50% in 22.2%, and < 1% in 19.4%. In terms of efficacy, the ORR of the induction chemo-IO regimen reached 86.11%, with a median PFS of 23.67 months and a median OS of 32.07 months. Treatment-related adverse events (TRAEs) of any grade occurred in 58.33% of patients, with pneumonitis being the most common (38.9%). Grade 3–4 pneumonitis was observed in 28.5% (4/11) of patients. Andrea Riccardo Filippi *et al.* proposed a de-intensified, hypofractionated thoracic radiotherapy strategy following induction chemo-immunotherapy, combined with concurrent durvalumab and maintenance durvalumab. However, the results of this study have yet to be reported [16]. These studies aimed to reduce the risk of recurrence and improve therapeutic efficacy through the early introduction of immunotherapy, achieving encouraging results in short-term local control. Nevertheless, further studies are needed in clinical practice to identify the most appropriate patient populations.

In the present case, after completion of induction chemotherapy combined with immunotherapy, the patient maintained good physical status (KPS score of 80) despite experiencing repeated grade III myelosuppression (mainly neutropenia and thrombocytopenia) during chemotherapy. Owing to concerns about the toxicity of CRT, the patient explicitly refused concurrent chemoradiotherapy. After a multidisciplinary discussion, considering that the patient was a 66-year-old male with a long-term smoking history and limited bone marrow reserve, he was deemed a high-risk candidate for CRT. To balance toxicity control with local disease control, we reviewed the literature. We found that Endostar, through inhibition of the VEGF signalling pathway, exerts multi-target anti-angiogenic effects, promotes tumour vascular normalization, and improves radiosensitivity. Therefore, Endostar was combined with radiotherapy in this case.

It is well known that treatment options for SCC remain relatively limited. Before initiating individualized treatment, peripheral blood genetic testing was completed, revealing PD-L1 expression of 20%. There is a lack of clinical research on combining ICIs and Endostar for locally advanced NSCLC. However, case reports have suggested that envafolimab combined with Endostar in patients with low PD-L1 expression advanced NSCLC significantly reduced both primary and metastatic lesions after 17 months of follow-up, with no obvious adverse events observed [17].

In clinical practice, Endostar has been shown to prolong survival in patients with squamous NSCLC without increasing the risk of bleeding. It is currently the only anti-angiogenic drug approved for first-line treatment of advanced squamous NSCLC. Compared with other anti-angiogenic agents, Endostar has a broader range of targets, covering VEGF-independent angiogenesis pathways. Its continuous intravenous infusion is associated with fewer side effects and higher safety. Therefore, during the maintenance treatment phase, the patient received pembrolizumab combined with Endostar, aiming to prolong disease control time through dual mechanisms of immune checkpoint blockade and vascular microenvironment modulation. Notably, although the left-sided lesion did not undergo surgery or local radiotherapy, the patient achieved a long PFS of 62 months through long-term targeted and immunotherapy, comparable to the expected efficacy reported in the PACIFIC-6 study. Previous studies have indicated that many patients develop innate or acquired resistance to immunotherapy, primarily owing to immune editing during the treatment process. This reduces tumor immunogenicity and re-establishing an immunosuppressive [18, 19]. Therefore, combining immunotherapy with other therapeutic strategies is crucial for improving survival outcomes. Regarding the immunotherapy response characteristics of SCC, PD-L1 expression is a core predictive biomarker of therapeutic efficacy. Patients with high PD-L1 expression (tumor proportion score ≥ 50%) demonstrate significantly improved ORR to immune checkpoint inhibitor monotherapy. Clinical data suggest that approximately 23% of patients with advanced NSCLC have tumors expressing PD-L1 at ≥ 50%, and this population is prioritized for treatment with PD-1/PD-L1 inhibitors.

Tumor mutational burden (TMB) is another important biomarker; patients with a high TMB (≥ 10 mutations/Mb) tend to exhibit better responses to immunotherapy, independent of PD-L1 status, and often achieve more favorable survival outcomes. Microsatellite instability (MSI) also plays a role—patients with high microsatellite instability (MSI-H) show significantly greater sensitivity to immunotherapy compared with those with low MSI (MSI-L) or microsatellite stable (MSS) tumors. Additionally, tumor-infiltrating lymphocytes (TILs), particularly high densities of CD8 + T cells within the tumor microenvironment, are associated with a 30% improvement in immunotherapy response rates, indicating that TILs may serve as a dynamic biomarker for therapeutic monitoring. In addition, testing for EGFR, ALK, and KRAS mutations helps to achieve accurate molecular subtype classification, comprehensively assess tumor characteristics, and predict tumor malignancy and prognosis. In treatment, EGFR-TKIs benefit patients with EGFR-sensitive mutations. ALK inhibitors are effective for ALK-positive patients.

Previous literature has shown that recombinant human endostatin (Endostar) is an anti-angiogenic agent that can normalize tumor microvasculature, reduce hypoxia, and enhance the efficacy of radiotherapy and chemotherapy. Endostar also promotes CD8 + T-cell infiltration into tumors, thereby contributing to an immune-supportive microenvironment [20]. Furthermore, Endostar can inhibit VEGF signalling, decrease the number of immunosuppressive cells such as myeloid-derived suppressor cells (MDSCs) and M2-type tumor-associated macrophages (TAMs), and increase mature dendritic cells and M1-type TAMs, effectively converting an immunosuppressive tumor microenvironment into one that supports immune activation [20]. A single-arm, multicenter, prospective phase II clinical trial conducted by Yuan S *et al*. demonstrated that the combination of envafolimab, Endostar, and chemotherapy in patients with advanced squamous NSCLC achieved an ORR of 81% (95% CI 58.1–94.6) and a disease control rate (DCR) of 100% (95% CI 83.9–100), indicating robust clinical activity and acceptable safety [21]. In a retrospective study, Wu L *et al*. evaluated the combination of Endostar with camrelizumab and chemotherapy in patients with advanced (stage IIIB and IV) NSCLC, reporting favorable efficacy and tolerability [22]. The HELPER study showed that, in patients with unresectable stage III NSCLC, continuous intravenous infusion of Endostar in combination with concurrent etoposide-platinum (EP) chemotherapy and radiotherapy did not significantly prolong median PFS but yielded improvements in OS, 2-year PFS rates, and exhibited a tolerable toxicity profile [23]. Given the therapeutic challenges in managing locally advanced SCC, the rational integration of Endostar, PD-1/PD-L1 inhibitors, radiotherapy, and chemotherapy is worthy of further exploration.

Conclusion

This paper provides a detailed report on the long-term follow-up of a patient with sMPLC following comprehensive treatment. The results indicate that for elderly patients with poor tolerance, sequential combination therapy using multiple treatment modalities can be employed. Early detection, histological/molecular confirmation, and multimodal management are crucial for improving long-term survival rates and quality of life in sMPLC patients. Among these, induction immunotherapy holds promise for enhancing local control rates, while the application of targeted therapy combined with radiotherapy and immunotherapy may re-activate patients' own immune responses, leading to effective survival benefits. Bone marrow suppression is the most common toxic reaction after chemotherapy and can significantly reduce patient compliance with treatment. Thyroid dysfunction, a typical manifestation of immune-related adverse events (irAE), often accompanies immunotherapy and radiotherapy and requires timely intervention by a professional team to maximize patient quality of life. In summary, clinical practice should be based on multidisciplinary consultations, integrating clinical information, imaging data, and histopathological/molecular characteristics to develop personalized clinical management strategies and treatment plans, thereby improving patients' long-term survival rates and quality of life.

Limitations

There are limitations in case reports, such as insufficient representativeness, lack of statistical significance, interference by confounding factors, and short follow-up time, which affect the promotion and long-term evaluation of research results. Prospective studies are needed in the future to confirm the important role of personalized therapy in clinical practice.

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Author contributions
Conceptualization: W.L. Investigation: J.Z. Data curation: Y.Y. and L.Z. Writing—original draft: Y.Y. and X.Z. Writing—review and editing: S.M. and W.L. Supervision: L.Z. Project administration: S.M.

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Data availability
The original data presented in the study are included in the article/supplementary material. Further inquiries can be directed to the corresponding author.

Declarations

Ethical approval and consent to participate.
This study was approved by the Shenzhen People's hospital institutional and research ethics committee.

Consent for publication
Written informed consent was obtained from the patient for publication of this case report and any accompanying images. A copy of the written consent is available for review by the Editor-in-Chief of this journal.

Competing interest
The authors declare that there are no conflicts of interest regarding the publication of this paper.

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